

STORMDOM

Global Storm Tracker

Home

Indian Ocean

West Pacific

South Pacific

Central Pacific

East Pacific

Atlantic

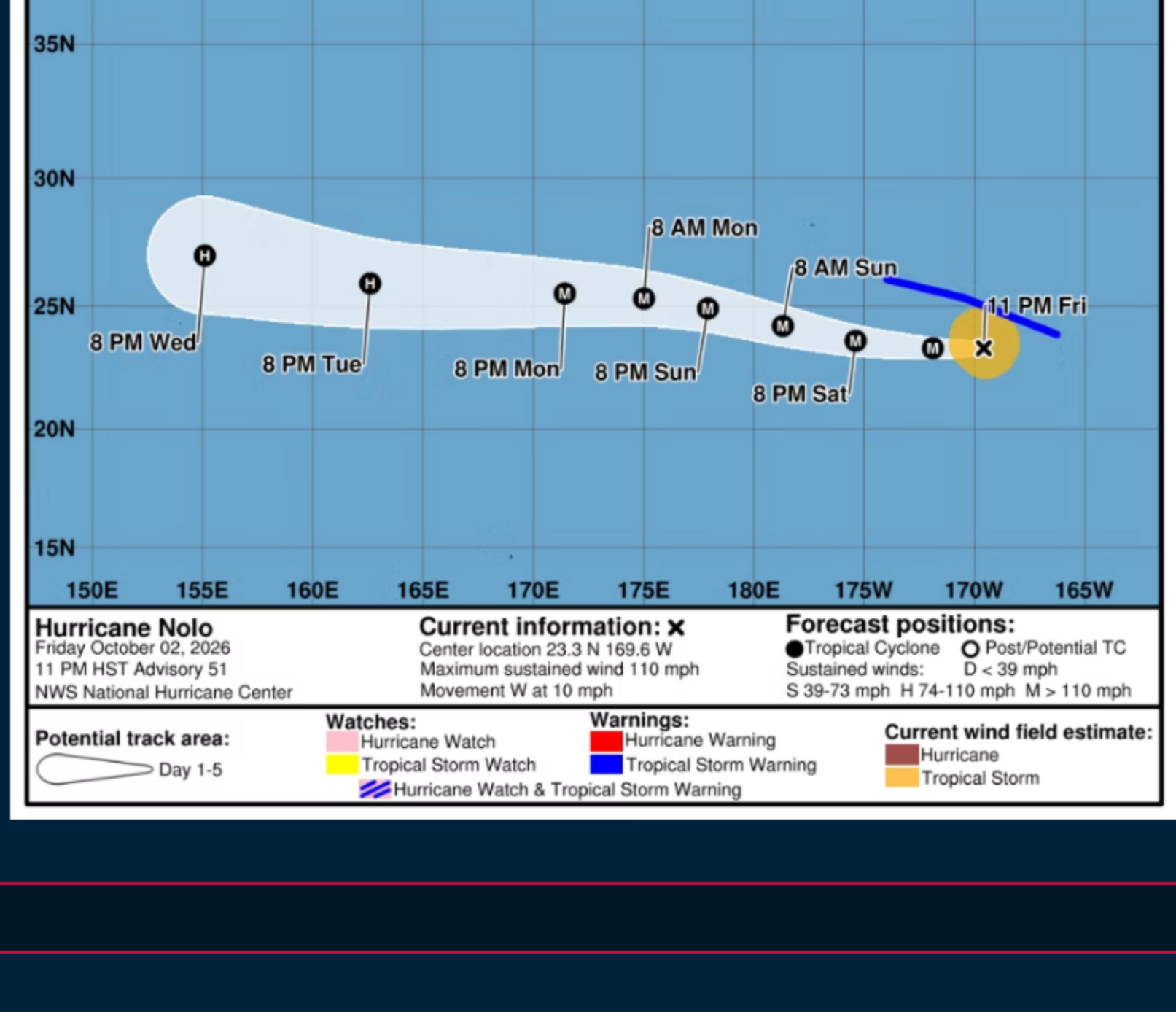
## Hurricane NOLO Tracker

Hurricane Nolo Advisory Number 51  
NWS Central Pacific Hurricane Center Honolulu HI EP15  
Issued by NWS National Hurricane Center Miami FL  
1100 PM HST Fri Oct 02 2026

...NOLO RAPIDLY INTENSIFYING WHILE MOVING WESTWARD...

SUMMARY OF 1100 PM HST...0900 UTC...INFORMATION

LOCATION...23.3N 169.6W  
ABOUT 150 MI...240 KM SSE OF MARO REEF  
ABOUT 215 MI...345 KM W OF FRENCH FRIGATE SHOALS  
MAXIMUM SUSTAINED WINDS...110 MPH...175 KM/H  
PRESENT MOVEMENT...W OR 270 DEGREES AT 5 MPH...17 KM/H  
MINIMUM CENTRAL PRESSURE...964 MB...28.47 INCHES



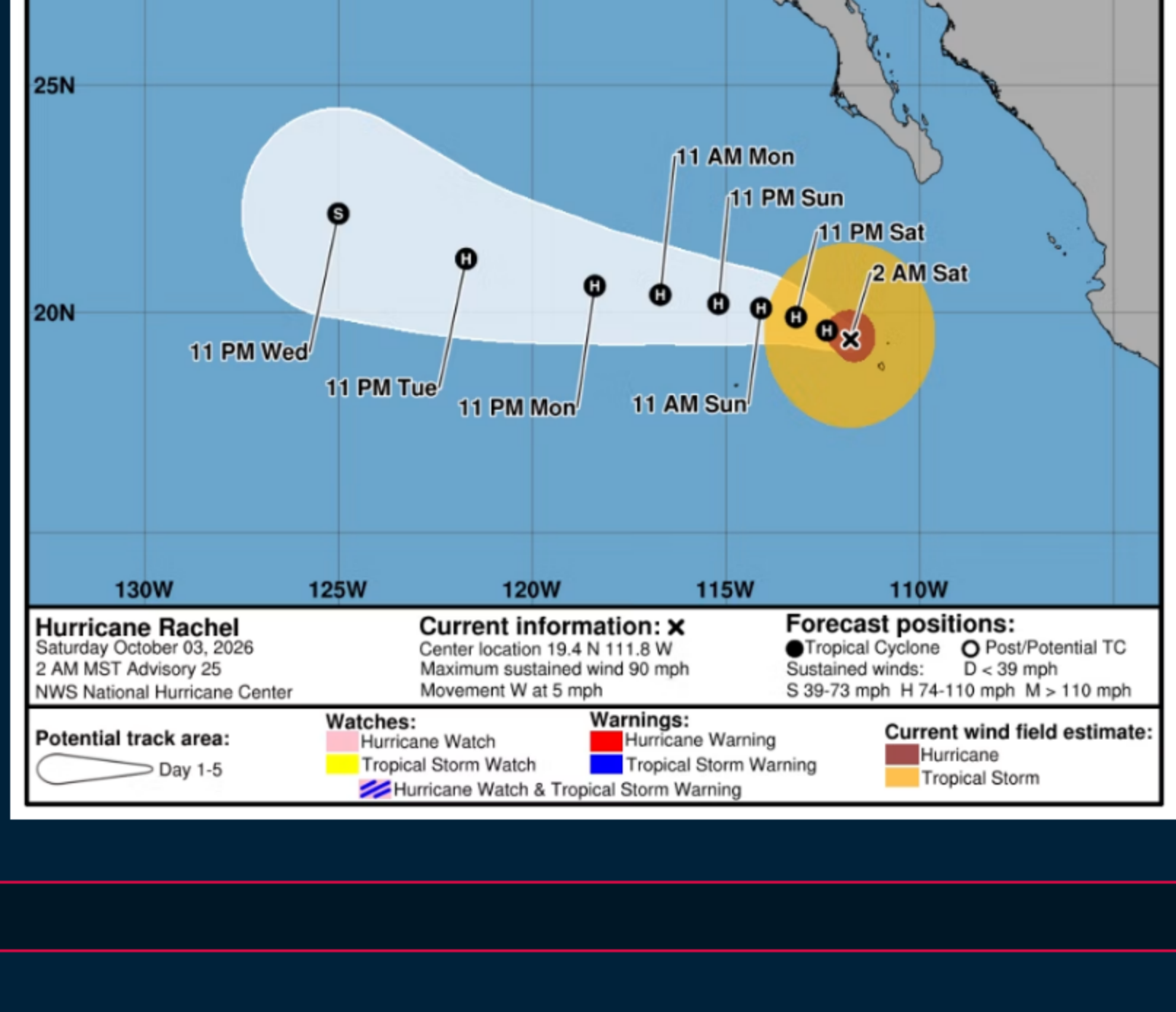
## Hurricane RACHEL Tracker

Hurricane Rachel Advisory Number 25  
NWS National Hurricane Center Miami FL EP182026  
200 AM MST Sat Oct 03 2026

...RACHEL WEAKENS SLIGHTLY AS IT CONTINUES WESTWARD...

SUMMARY OF 200 AM MST...0900 UTC...INFORMATION

LOCATION...19.4N 111.8W  
ABOUT 270 MI...435 KM SSW OF THE SOUTHERN TIP OF BAJA C  
MAXIMUM SUSTAINED WINDS...90 MPH...150 KM/H  
PRESENT MOVEMENT...W OR 270 DEGREES AT 5 MPH...7 KM/H  
MINIMUM CENTRAL PRESSURE...969 MB...28.62 INCHES



## Typhoon CHOI-WAN Tracker

1. TYPHOON 26W (CHOI-WAN) WARNING NR 014  
01 ACTIVE TROPICAL CYCLONE IN NORTHWESTPAC  
81 SUSTAINED WINDS BASED ON ONE-MINUTE AVERAGE  
WIND RADII VALID OVER OPEN WATER ONLY

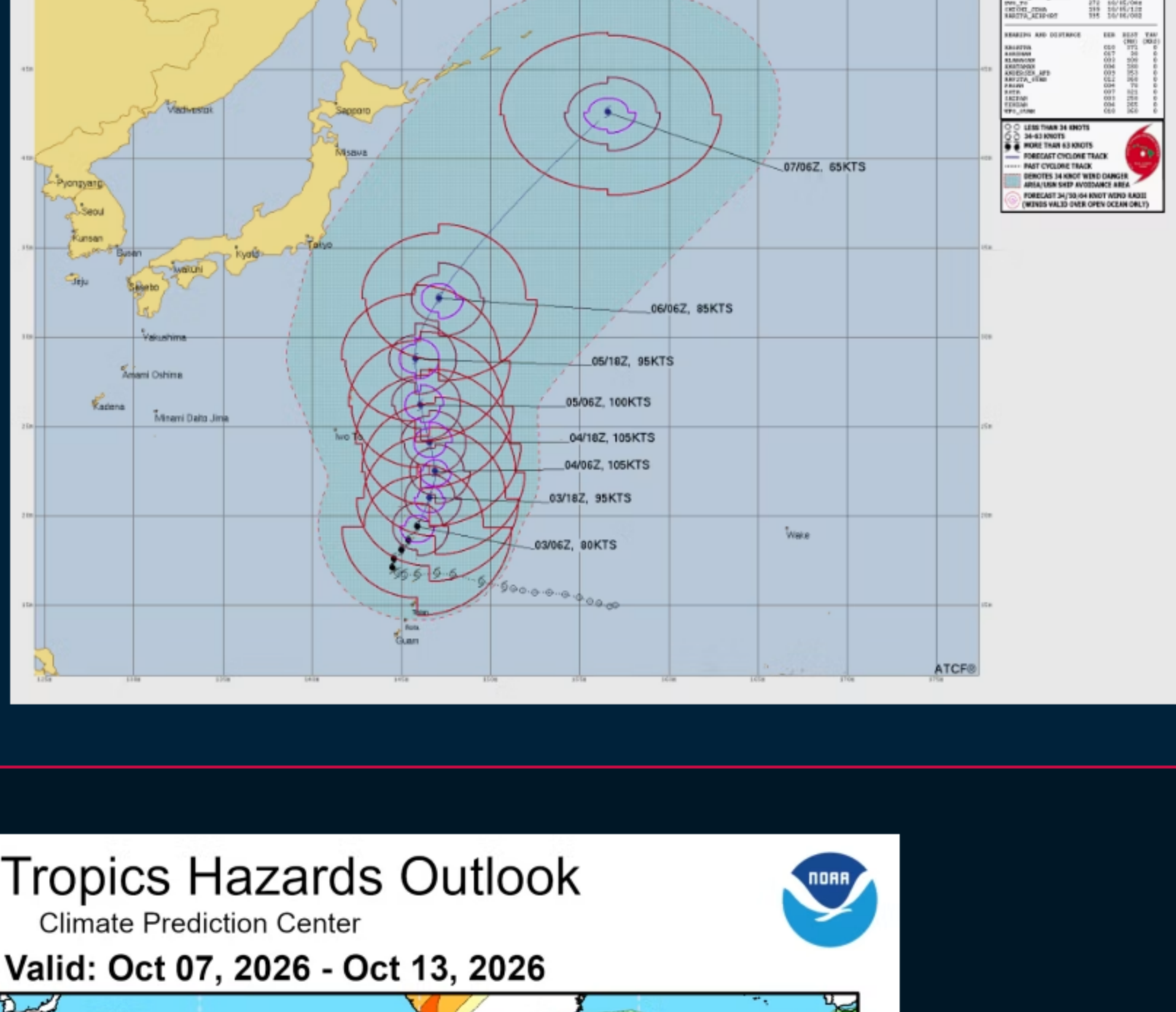
WARNING POSITION:  
030600Z --- NEAR 19.4N 145.9E  
MOVEMENT PAST SIX HOURS - 030 DEGREES AT 09 KTS  
POSITION ACCURATE TO WITHIN 030 NM  
POSITION BASED ON CENTER LOCATED BY SATELLITE  
PRESENT WIND DISTRIBUTION:  
MAX SUSTAINED WINDS - 080 KT, GUSTS 100 KT  
WIND RADII VALID OVER OPEN WATER ONLY

RADIUS OF 064 KT WINDS - 040 NM NORTHEAST QUADRANT  
055 NM SOUTHEAST QUADRANT  
055 NM SOUTHWEST QUADRANT  
040 NM NORTHWEST QUADRANT

RADIUS OF 050 KT WINDS - 080 NM NORTHEAST QUADRANT  
100 NM SOUTHEAST QUADRANT  
080 NM SOUTHWEST QUADRANT  
080 NM NORTHWEST QUADRANT

RADIUS OF 034 KT WINDS - 210 NM NORTHEAST QUADRANT  
300 NM SOUTHEAST QUADRANT  
240 NM SOUTHWEST QUADRANT  
170 NM NORTHWEST QUADRANT

REPEAT POSIT: 19.4N 145.9E



Global Tropics Hazards Outlook

Climate Prediction Center

Week 2 - Valid: Oct 07, 2026 - Oct 13, 2026

Week 3 - Valid: Oct 14, 2026 - Oct 20, 2026

Tropical Cyclone (TC) Formation Probability

>20% >40% >60%

Above-Average Rainfall Probability

>50% >65% >80%

Below-Average Rainfall Probability

>50% >65% >80%

Above-Average Temperatures Probability

>50% >65% >80%

Below-Average Temperatures Probability

>50% >65% >80%

Issued: 09/29/2026  
Forecaster: Barandiaran

Graphic provided by Climate Prediction Center

## See my new article: [How I Met Hurricane Camille](#)

## What Is a Hurricane?

A hurricane (or typhoon, or severe tropical cyclone), the strongest storm on Earth, is a cyclonic (rotary) storm that derives its energy from cloud formation and rainfall, unlike frontal cyclones that derive their power from a temperature gradient.

A hurricane begins as a tropical depression with a sustained wind speed of less than 39 mph (35 knots; 63 km/hr). As the system strengthens, it becomes a tropical storm with winds from 39 to 73 mph (35-63 knots; 63-118 km/hr). Tropical storms are named in the Atlantic, East, Central and Northwest Pacific, in the South Indian Ocean, and in the Arabian Sea. When the winds are sustained (based on a one-minute average) at 74 mph (64 knots; 119 km/hr), the storm becomes: In the Atlantic Ocean, East Pacific, Central Pacific (east of the International Dateline) and Southeast Pacific (east of 160°E) a Hurricane; in the Northwest Pacific (west of the International Dateline) a Typhoon; in the Southwest Pacific (west of 160°E) and Southeast Indian Ocean (east of 90°E) a Severe Tropical Cyclone; in the North Indian Ocean a Severe Cyclonic Storm; and in the Southwest Indian Ocean (west of 90°E) a Tropical Cyclone.

## The Saffir-Simpson Hurricane Scale

Category 1 – 64-82 knots (74-95 mph; 119-153 km/h). Damage is limited to foliage, signage, unanchored boats and mobile homes. There is no significant damage to buildings. The main threat to life and property may be flooding from heavy rains.

Category 2 – 83-95 knots (96-110 mph; 154-177 km/h). Roof damage to buildings. Doors and windows damaged. Mobile homes severely damaged. Piers damaged by storm surge. Some trees blown down, more extensive limb damage.

Category 3 – 96-112 knots (111-129 mph; 178-208 km/h). Major Hurricane. Structural damage to some buildings. Mobile homes completely destroyed. Roof damage is common. Storm surge begins to cause significant damage in beaches and harbors, with small buildings destroyed.

Category 4 – 113-136 knots (130-156 mph; 209-251 km/h). Structural failure of some buildings. Complete roof failures on many buildings. Extreme storm surge damage and flooding. Severe coastal erosion, with permanent changes to the coastal landscape not unheard of. Hurricane force winds extend well inland.

Category 5 – 137+ knots (157+ mph; 252+ km/h). Complete roof failure on most buildings. Many buildings destroyed, or structurally damaged beyond repair. Catastrophic storm surge damage. In the Northwest Pacific, a typhoon that reaches 150 mph (241 km/hr) is called a Super Typhoon.

|               | SAFFIR-SIMPSON SCALE |         |         |              |
|---------------|----------------------|---------|---------|--------------|
| Category      | Knots                | MPH     | KM/H    | Damage       |
| 1             | 64-82                | 74-95   | 119-153 | Minimal      |
| 2             | 83-95                | 96-110  | 154-177 | Moderate     |
| 3             | 96-112               | 111-129 | 178-208 | Extensive    |
| 4             | 113-136              | 130-156 | 209-251 | Extreme      |
| Super Typhoon | 130+                 | 150+    | 241+    | Catastrophic |
| 5             | 137+                 | 157+    | 252+    | Catastrophic |

## Storm Surge

Historically, storm surge is the primary killer in hurricanes. The exact storm surge in any given area will be determined by how quickly the water depth increases offshore. In deep-water environments, such as the Hawaiian islands, storm surge will be enhanced by the rapidly decreasing ocean depth as the wind-driven surge approaches the coast. The peak storm surge is on the right-front quadrant (left-front in the Southern Hemisphere) of the eyewall at landfall, where on-shore winds are the strongest, and at the leading edge of the eyewall. Contrary to a popular myth, the storm surge is entirely wind-driven water—it is not caused by the low pressure of the eye. Another factor in the severity of the storm surge is tide. Obviously, an 18-foot storm surge at high tide is that much worse than an 18-foot surge at low tide.

## Tropical Cyclone Formation

By Jonathan Edwards

Tropical Cyclone Genesis is the technical term for the process of storm formation that leads ultimately to what are called hurricanes, typhoons, or tropical cyclones in various parts of the world.

This occurs when, in the Northern Hemisphere, the Intertropical Convergence Zone, or ITCZ, shifts northward out of the doldrums and atmospheric conditions become favorable for tropical cyclone formation after about the middle of May.

A series of low-pressure ripples develops within the ITCZ. These are known as tropical waves and progress from east to west. In the late season, they typically shift their movement toward the west-northwest, or even northwest, after crossing 45° or 50° W longitude.

These tropical waves, ideally embedded in the deep-layer easterly flow, contain a northeast wind shift. This is typically referred to as a “convergence”, where lines of equal atmospheric pressure are pressed together between the high-pressure ridge to the north and the developing low-pressure systems. The divergence that results ahead of the convergence zone gives us a northeasterly wind as the axis of the tropical wave approaches. Gusts up to 25 mph may occur. Sometimes there can be gusts to tropical storm force in stronger waves. There can be next to no weather associated with these waves, and they may pass virtually unnoticed. More typically, there are bands of disturbed weather riding the axis of the wave.



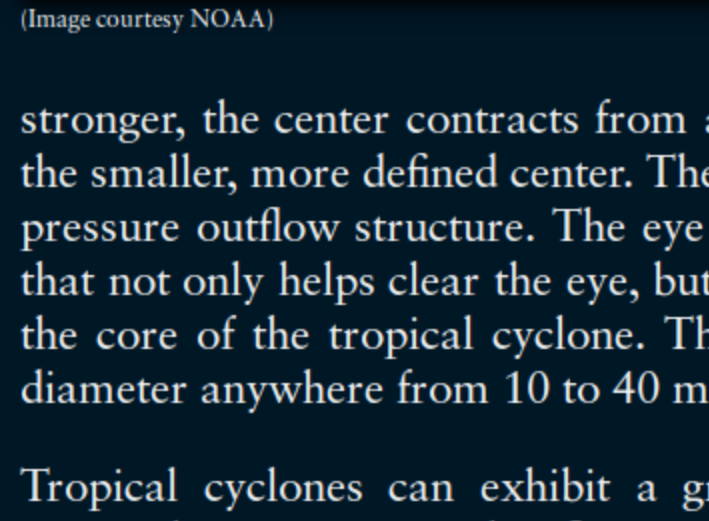
(Graphic by Robert Simmon, NASA GSFC)

When the wave passes over warmer waters (SSTs), convection and resulting rainfall are enhanced. This greater rainfall is concomitant with falling surface pressures. By the time these pressures fall to 1008mb, it is likely that the northeast wind has closed off to a southwest wind on the backside of the wave. The forward motion of the wave completes the closure on the northern side of a broad low-level center, and a tropical depression has formed.

We often hear that a tropical depression has formed, but conditions are unfavorable for further development. There are two conditions that must be present for the tropical depression to continue its development: warm SSTs (above 79° Fahrenheit/26° Celsius) and low vertical shear. A tropical cyclone derives its power from the warm waters below. In addition, a strong anticyclone directly above the low-level inflow is favorable. As a tropical cyclone is pulling in warm, moist air at the surface, it must also evacuate this inbound flow aloft. This occurs in the upper levels of the atmosphere, where high pressure facilitates the development of the cyclone by evacuating the flow from the lower levels of the cyclone. Every powerful hurricane has an equally powerful high pressure system over it. At the surface, the air spirals inward in a counter-clockwise direction, and spirals out at the top in a clockwise direction. In the Southern Hemisphere, it is reversed: clockwise inbound, counterclockwise outbound.

If the upper-level high pressure system does not remain over the cyclone, it means there is shear ahead. This is a strong jet of air that is blowing directly over the cyclone, ripping the cloud tops off the deep convection. This has the effect of breaking down the whole mechanism. This is known as vertical shear. Vertical shear usually comes from a westerly or southerly direction, and can occur if the cyclone is located in an unfavorable position near a cold front or upper-level low pressure system. Another factor that can interfere with the development of a tropical cyclone is subsidence. Subsidence is the sinking of air. This happens on the edge of an upper-level high-pressure system. Subsiding air has the effect of suppressing thunderstorm formation. This also is why a tropical storm that tries to form near an established hurricane has a very difficult time—the cyclone is on the edge of the hurricane's upper-level outflow, and may have to contend with both subsidence and shear. The effect of shear on a cyclone can range from a failure to develop further to catastrophic collapse of the tropical cyclone's support structure.

Tropical cyclones have the low-level circulation and the upper-level circulation (outflow), whose formation was discussed above. There is also a mid-level circulation. The mid-level circulation is similar in structure to the low-level circulation, and is critical to the survival of a tropical cyclone that is passing over land. The low-level circulation can be severely disrupted, or even dissipated, by interaction with land, especially mountainous terrain. If the mid-level circulation remains intact, the cyclone can regenerate rapidly when it reemerges over water, providing other factors are favorable.



(Image courtesy NOAA)

Assuming all the ingredients are in place—warm SSTs, upper-level high pressure, and falling surface pressures—the cyclone will develop and reach a point of rapid intensification. It is one of nature's perfect machines. As warm waters feed the convection swirling around the center, heavy rainfall lowers surface pressures, high pressure aloft evacuates the inflow, which intensifies the inflow of warm, moist air, which in turn increases the rainfall and brings about a more rapid fall in central pressure.

Eye formation begins when a tropical storm reaches approximately 65mph, provided conditions are favorable for strengthening to continue. The eyewall begins to make its appearance, usually on the eastern (Northern Hemisphere) edge of the center. As the system becomes better organized and the eyewall narrows, the eye begins to form. The eye is a calm area in the center of the storm. The eyewall creates subsidence that not only helps clear the eye, but can also produce a clear spot in the center, which is an area of relatively weak convection outside the core of the tropical cyclone. The combination of eye and moat can make the storm's core look like a doughnut. The eye can have a diameter anywhere from 10 to 40 miles across.

Tropical cyclones can exhibit a great deal of durability provided that the upper level support remains and the southerly (Northern Hemisphere), or tropical, inflow is present. Here are some examples: In 1998, Hurricane Mitch developed into a supermassive Category 5 hurricane. Nevertheless weakening began when the center moved to a position directly north of Honduras, cutting off the southerly inflow, even though the eye was still over a hundred miles offshore. Later, Mitch maintained its mid-level core against all odds over the mountains of Central America because he was able to advect moisture from the East Pacific. Reintensification to tropical storm strength was almost immediate after reaching the Bay of Campeche. In 1988 Gilbert hit the Yucatan near Cancun as a Category 5 hurricane. The Yucatan peninsula, though flat, extended far enough south through the critical southerly inflow zone that Gilbert never recovered, even after moving over the open waters of the western Gulf of Mexico. Contrast this with the northern Gulf coast and numerous examples from Camille to Elena and beyond, where the proximity of the eye to land is not necessarily a favorable factor, although the extensive shelf water can sometimes be an inhibiting factor, if a slow-moving storm generates enough upwelling. A recent Southern Hemisphere example of this is Severe Tropical Cyclone Monica, which developed into a “super cyclone” just north of Australia. The northerly (equatorward) inflow over the Arafura Sea fueled this system.

What are the factors that contribute to the decay of a tropical cyclone? They are Upwelling, Entraining dry air, Moving over cool waters, Exposure to upper-level westerlies, and finally Landfall.

**Upwelling.** When a hurricane stalls, its movement is has fallen below 5mph, or its movement is erratic over a small area, the wave action caused by the strong surface winds churns the ocean surface and produces upwelling. This has the effect of cooling the temperature of the sea surface over an area 200 to 300 miles across. The result is weakening. It is possible for a hurricane to stall in one area long enough that it dissipates. In 2004, Hurricane Frances stalled off the southeast Florida coast long enough that the core of the hurricane collapsed. Frances eventually made landfall as a Category 1 hurricane, instead of the earlier expected Category 3.

**Entraining dry air.** Sometimes, during the peak season, when tropical cyclones approach continental land masses, they may entrain dry air as part of their interaction with the frontal troughs that carry cool, dry air behind them. The dry air kills off the convection that drive the hurricane's engine. If the dry air entrains deeply enough, the result can be dissipation.

**Moving over cool water.** Similar to upwelling, when a tropical cyclone moves over cool water (below 77° Fahrenheit/25° Celsius), it begins to weaken. Eventually this causes dissipation, particularly in the East Pacific. In the Atlantic, if the storm is caught in the mid-latitude westerlies and begins to recurve to the northeast, it generally becomes an extratropical storm by the time it has reached about 45° W. This is the so-called “graveyard” of Atlantic hurricanes. The storm becomes extratropical when it has transitioned to a cold-core, baroclinically driven system, and eventually becomes a hurricane in the north Atlantic, or is absorbed by a large scale, having maintained a recognizable inner core.

**Exposure to upper-level westerlies.** Frequently, the ridge of high pressure that keeps the hurricane heading toward the west erodes. In the Atlantic Ocean, this may result in strong upper-level westerlies diving down and impinging on the northern edge of the hurricane's upper-level support structure. Poleward outflow is restricted. A hurricane can still maintain itself with restricted in one quadrant. If outflow becomes too restricted, resulting shear can disrupt the inner core.

**After landfall.** Tropical cyclones cannot survive over land, once their access to warm SST is removed. A powerful hurricane, such as Hugo, which hit Charleston, S.C., in 1989 can project life-threatening hurricane force winds over two hundred miles inland. As the storm progresses inland, it can dump a huge amount of rain—sometimes measured in feet. The storm may evolve into a frontal cyclone that continues to cause widespread damage. The best example of this is Hurricane Camille in 1969—the strongest hurricane ever to make landfall on the continental United States with winds sustained at 190 mph and gusts well exceeding 200 mph—which roared up the Mississippi Valley and eventually exited off the East Coast. Camille maintained tropical storm strength as far as Memphis, Tennessee. Most hurricanes will diminish in strength rapidly after landfall, reaching tropical depression strength by 48 to 72 hours. The main threat from the dying storm is from tornadoes and inland flooding. The right-hand quadrant of a hurricane or typhoon (in the Northern Hemisphere, left-hand in the Southern Hemisphere) is most frequently the strong side of the storm. This is because the forward motion is added to the counter-clockwise punch of the storm. Thus, the right-hand side of the hurricane contains the strong on-shore flow. This is where the maximum storm surge flooding, and the greatest potential loss of life, will be experienced. After landfall, the circulation's motion over land causes a great deal of turbulence, which results in tornadoes. These are especially likely in the forward right-hand side of the storm's path.